DRAFT VERSION

(work in progress)

Sample-Ama Uris

TABLE OF CONTENTS

Data Backup Plan Portfolio Project by Ama Jessica /Uris

1. Context & Scope	01
2. Roles and Responsibilities	02
3. Backup Strategy	03
4. Risk management	04
5. Conclusion	05

Sample-Ama Uris

Data Backup Plan

1. Context

Imagine that you're working in the IT department for a large retail franchise and have a team of 20 experts. Your supervisor tasks you with developing a backup plan.

Identify the person, team, or device that will complete backups. For this activity, you have three choices:

- Store manager
- IT team
- Automated backup system

1.1 Scope

All the critical data a large retail franchise will back up includes five types of proprietary and private data listed below:

- **Private & Proprietary** Employee Data
    - Employee records, i.e. employment contracts, contact and payroll information.
- **Private** Customer Data
    - I.e. contact information, feedback and support records, and service history.
- **Proprietary** Financial Data
    - Financial records, i.e. invoices, receipts, ledgers — accounting and tax records.
- **Proprietary** Company Data
    - I.e. sales reports, production and inventory records.
    - Intellectual property, i.e. patents, copyrights and trademarks.
- **Proprietary** System Data
    - I.e. software license, application backups, and system images and configurations.

Sample-Ama Uris

Furthermore, our confidentiality, integrity, and availability is maintained by recovery processes. Consequently, all data backups are guarded by various forms of asymmetric encryption including file encryption, network encryption, email encryption and full-drive encryption. Additionally, aligned with our data loss prevention systems, network encryption will confirm authentication, integrity and non-repudiation, maintaining all elements of the CIA triad. The future investment of what comes with good security are reputational, operational and financial.

Sample-Ama Uris

2. Roles and Responsibilities

The **automated backup system** will complete each backup and the time it is performed will positively affect business operations and productivity.

Multiple assigned **IT team members** and their **Compliance Officer** will have primary responsibility over the automated system. This includes consistently scheduled monitoring, testing, and performance spot-checks on the system to ensure the backup maintains accuracy and validity. Each member of the team with this expertise will assume this responsibility on a rotary basis, the details of which will be provided for in the procedures that will follow this Backup Plan. The IT team will also need to record and consolidate these activities into a monthly or weekly (depending on capacity, resources, and risk level) report. The document will be sent to relevant stakeholders such as the store manager, the IT manager, the IT department's **Compliance officer**, and the **Chief Information Security Officer** (CISO).

Alongside the **store manager**, the **compliance officer** will have greater monitoring & reporting responsibilities. The store manager, IT manager, and compliance officer will be responsible for storing a record of the monthly or weekly system reports. These reports can be sourced for other internal reports, executive meetings, decision-making and internal audits.

Sample-Arma Uris

3. Backup Strategy

Backup type required

Due to the large size of the retail franchise, a **differential** backup system would enhance the efficiency of business processes and align with cybersecurity best practices by:

- Creating a secure environment for us all to operate in. Our people are protected from data loss, and we extend that protection to our clients and consumers, whose data is equally private.
- Ensuring availability. Both proprietary and private data are governed by regulations that require keeping data **available** through timely restoration (HIPAA).
- Improving disaster recovery **timelines** faster than full backups.
- Maintaining an **open dialogue** with our clients/customers/consumers (end users), executives, management and employees at all levels of the organisation. Using the discipline of change management to guide our policy and procedures with inclusion and creative problem-solving while considering resources and business processes.
- Complying with **industry standards** that are informed by national, regional, and global regulations and law.

Backup storage type and location

Using best practices to guide our policy, we use a **3-2-1** storage strategy that necessitates the following:

- Create **3** backups of all **critical data**: **1** primary and **2** additional.
- Use **2** different **storage types** (i.e. cloud, hard disk drives, tapes).
- Keep at least **1** backup **offsite**.

As such, our storage policy reflects this strategy by:

- Creating **3 critical data backups**:
    - Storing **1** backup **offsite**: On tape that is kept off the premises.
    - Storing **2** backups **onsite**: On the cloud, and on hard disk or optical drives, on the premises.

Sample-Ama Uris

Backup schedule

The backup schedule is designed based on: available resources, the company's risk tolerance, data classification, significance to daily operations, the frequency of changes made, and the time & resources available to perform it.

Consequently, we maintain the following backup schedule:

- **Full** backups **every midnight** with **differential** backups **throughout the day**, using a time span that considers how it affects stakeholders, and how much it may slow the system down.
- This backup schedule is open to change according to resources and time availability.

Notes on Communication

A. An excerpt of internal communication content featured in the open dialogue, increasing awareness using a creative publication for different levels of employees, customers and other relevant stakeholders:

To the kids that use the technology.

We're working with daily differential backups and weekly full backups for improved downtime and better disaster recovery efforts.

To the kids that watch the technology.

We have a data loss prevention system that makes use of two types of data backup methods; both save us downtime and less damage = less vulnerable to threats.

To the kids that watch users and other watchers.

Our data loss prevention policy is meeting the requirements of the following list of regulators:

- Namibia
    - Cybercrime Bill 2026 (Incoming).
    - NAM-CSIRT's Incident Management Guidelines.
    - Data Protection Bill 2026 (Incoming).

Sample-Ama Uris

- S.A
    - Protection of Personal Information Act (POPIA).
- Nigeria
    - Nigeria Data Protection Act (NDPA).
- Kenya & Mauritius
    - Office of the Data Protection Commissioner (ODPC).
- U.S
    - Health Insurance Portability and Accountability Act (HIPAA).
    - Gramm-Leach-Bliley Act (GLBA) safeguards.
- E.U
    - General Data Protection Regulation (GDPR).
    - Digital Operational Resilience Act (DORA).
- Industry
    - National Institute of Standards and Technology (NIST SP 800-series).
    - Payment Card Industry Data Security Standard (PCI-DSS).
    - Internet Engineering Task Force (IETF) Internet Protocols.
    - The International Society of Automation (ISA) and the International Electrotechnical Commission (IEC)
        - ISA/IEC 62443 (retail environments that use operational technology).
    - The American Institute of Certified Public Accountants (AICPA)
        - SOC (System and Organisation Controls) 2 Type I & II audits.

To the kids that buy the technology.

We are following the most restrictive regulation with a core defence against loss, downtime and expenses.

To the technology the kids access, use and protect. *(encrypted)*

1000111010101010010001110001100001010010010101010 01.

B. Share the Data Backup Plan and other important and relevant policy publications using *'The kids that…'* cover titles for each '*kid'* category mentioned above. This will provide invested financial, operational and reputational benefits for the organisation.

Sample Ama Uris

4. Risk Management

Need to create a risk assessment and control matrix for the backup eco-system.

Try creating it here, your own. Understand what the technical control is, understand what each means in laymans terms.

Risk ID	Risk	Likelihood / Impact	Control & Mitigation Strategy

Sample-Ama Uns

5. Conclusion

We evaluate our data backup plan using comprehensive criteria that aligns with the Generally Accepted Privacy Principles (GAPP), the fundamental confidentiality, integrity and availability cybersecurity triad, and the three key elements of cybersecurity: people, process and technology.

Implementing this plan is a significant part of the bigger plan: preventing and mitigating financial, operational and reputational damage to the organisation and its stakeholders, and promoting a security culture that we all uphold and rely on.

In this case, our values and business is to protect all confidential data, including various types of proprietary data, and private employee and customer data.

Considering various factors that affect when, where and what type of backup is required, I have recommended a differential backup system that includes full backups daily: stored online and offline; two on site, and one off site.

As previously explained, our data backup plan maintains the confidentiality and availability of critical data, especially during and after a security event. Using primarily administrative and technical controls in this backup plan, we account for security at each step of the data lifecycle. We use modern asymmetric encryption algorithms and multiple data loss prevention systems within our Backup Strategy. This internal communication can be found in the Data Backup Plan and the data backup procedures (SOPs) that will follow. Further communication for the public (external stakeholders) will be provided through a high-level system availability SLA, data privacy and security policies, and/or business continuity summaries.

Sample-Ama Uris